Problem 5.8

Suppose you lend \$60 to your sister on September 14, at an annual rate of simple interest of 10%, to be repaid on December 25. How much does she have to pay you back? Use actual/360 time measurement.

We use the simple interest formula:

$$I = Prt,$$

where $P = 60$, $r = 0.10$, and t is the time in years.

Number of Days

Using actual time from September 14 to December 25:

$$16 + 31 + 30 + 25 = 102 \text{ days.}$$

Since the problem asks for actual/360 time measurement, we use the exact number of days over a 360-day year:

$$I = 60(0.10)\left(\frac{102}{360}\right).$$

Therefore,

$$I = 1.70.$$

The total amount she must repay is principal plus interest:

$$60 + 1.70 = 61.70.$$

So, she has to pay back

$$\boxed{\$61.70}.$$